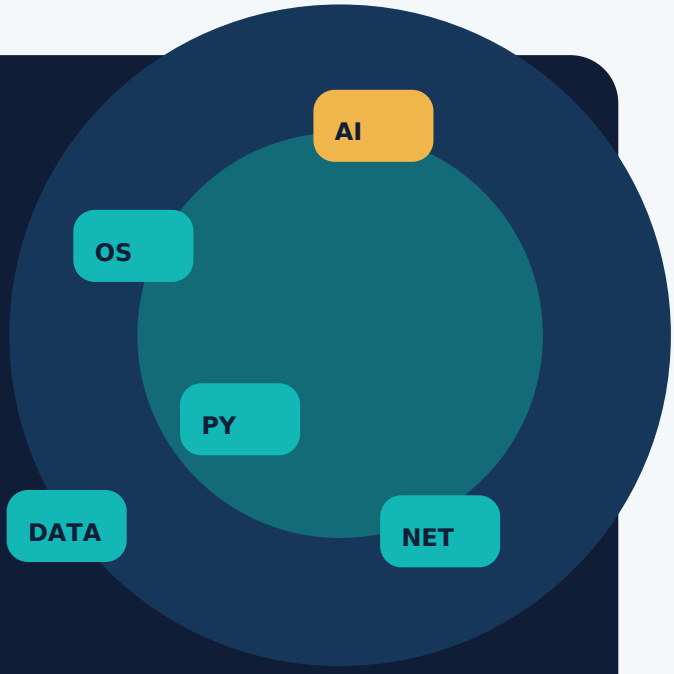
CLASS 10 + CLASS 12

SELECTED VISUAL VIDEO TOPICS

Faceless Shorts first. Full lectures later.



PURPOSE

A short, selective production list for social-first animations, challenges and micro-explainers. Every idea is labeled with the exact class, book, chapter and a thumbnail-ready phrase.

61	28	33
CURATED TOPICS	CLASS 10	CLASS 12

SOURCE: THE TWO UPLOADED PUNJAB COMPUTER SCIENCE AND ENTREPRENEURSHIP BOOKS

Note: creative video concepts and quiz examples are original, not copied textbook questions.

Produce these ideas BEFORE the sequential full-lecture series

1 This list is deliberately selective

These are the concepts likely to benefit from movement, comparisons, decision paths, simulations or an interactive challenge. It is NOT a repetition of every syllabus subtopic.

2 Each line is book-mapped

The header on every card identifies Class 10 or 12, the exact textbook name, its chapter title and the chapter starting page from the contents page. The hook and thumbnail language are editorial, not official textbook titles.

3 Make ONE master animation

Build a 16:9 original 2-5 minute faceless explainer when valuable. Export a distinct 9:16 Short from the best 25-50 seconds. Do not count reposts on five platforms as five unique topics.

4 Keep difficulty honest

For high-level analogies (networking, AI, statistics), label simplifications. Verify original quiz keys before release. Do not repurpose source textbook scans or reproduce its diagrams verbatim.

5 Publish an identity early

Use consistent voice, name, website and brand mark from the first week. A short on-camera hook can coexist with faceless animated explainers; avoid waiting months for all face-to-camera chapters.

Priority and effort legend

A = publish early / high visual appeal. B = strong follow-up. S = lightweight typography or code animation; M = process diagram or interactive animation; L = multiple scenes, custom diagrams or substantial verification. All effort labels are relative, not hour guarantees.

Short-video packaging

0-3 sec	Start with a question, misconception or surprise.
3-25 sec	Animate ONE mechanism. Keep labels readable on mobile.
25-45 sec	Let the viewer predict the outcome or compare choices.
Final 3 sec	Reveal the answer and invite practice on the website.

Only selected animated concepts, grouped by book

COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

01	Operating Systems: Structure and Services	5 visual topics | book p. 1
02	System Recovery and Advanced Maintenance	3 visual topics | book p. 20
03	Introduction to Python Programming	3 visual topics | book p. 40
04	Control Structures in Python	4 visual topics | book p. 55
05	Data Science	4 visual topics | book p. 69
06	Introduction To Artificial Intelligence (AI), and Machine Learning (ML)	4 visual topics | book p. 88
07	Applications of AI	3 visual topics | book p. 101
08	Digital Entrepreneurship	2 visual topics | book p. 115

COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

01	Computer Networks	6 visual topics | book p. 1
02	Computational Thinking & Algorithms	4 visual topics | book p. 20
03	Object Oriented Programming Using Python	3 visual topics | book p. 33
04	Development of Graphical User Interface (GUI)	3 visual topics | book p. 45
05	Code Testing and Debugging	3 visual topics | book p. 60
06	Data and Analysis	4 visual topics | book p. 72
07	Hypothesis Testing	3 visual topics | book p. 89
08	Applications of Computer Science	3 visual topics | book p. 103
09	Cybersecurity and Safe Digital Collaboration	4 visual topics | book p. 118

The 16 strongest first-post candidates across both classes

01	CLASS 10 / CH 1: What does an OS actually do?	M effort
02	CLASS 12 / CH 9: Spot the phishing message	M effort
03	CLASS 10 / CH 6: AI versus ML	M effort
04	CLASS 12 / CH 1: Network topologies	M effort
05	CLASS 10 / CH 3: Variables as labeled boxes	S effort
06	CLASS 12 / CH 2: Algorithm to flowchart	M effort
07	CLASS 10 / CH 1: The process life cycle	M effort
08	CLASS 12 / CH 3: Class versus object	M effort
09	CLASS 10 / CH 5: How a graph can mislead	M effort
10	CLASS 12 / CH 4: Your first GUI button	M effort
11	CLASS 10 / CH 4: For-loop versus while-loop	M effort
12	CLASS 12 / CH 1: OSI layers as a delivery story	L effort
13	CLASS 10 / CH 7: Speech recognition pipeline	M effort
14	CLASS 12 / CH 6: Train versus test split	L effort
15	CLASS 12 / CH 8: Smart agriculture with IoT	L effort
16	CLASS 12 / CH 5: Syntax vs runtime vs logic errors	M effort

Tip: The first 16 are a recommendation, not a claim about algorithmic performance. Rotate grades to learn whether Class 10 or Class 12 is responding. A 2-5 minute video can be optional for harder topics; start with an excellent Short where time is limited.

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 1: Operating Systems: Structure and Services

Textbook chapter begins at printed page 1 | Selected faceless concepts: 5

CLASS 10 / CH 1

C10-01-01

What does an OS actually do?

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Operating Systems: Structure and Services (p. 1)

HOOK	Who is controlling all your apps?
SHORT ANIMATION	Phone opens 3 apps; highlight CPU, RAM and OS as the invisible traffic controller.
EXTEND TO 2-5 MIN	4-minute journey of one app request from tap to hardware response.
QUIZ / WEBSITE	If two apps need the CPU, who coordinates access? Answer: the OS. Drag each task to CPU, memory or files and check the OS role.

THUMBNAIL: WHO CONTROLS YOUR PHONE?

A PRIORITY | M EFFORT

CLASS 10 / CH 1

C10-01-02

Kernel versus shell

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Operating Systems: Structure and Services (p. 1)

HOOK	You click a button. What happens next?
SHORT ANIMATION	Animate user click > shell or interface > kernel > hardware > feedback.
EXTEND TO 2-5 MIN	Explain kernel duties and the user-facing interface with a layered diagram.
QUIZ / WEBSITE	Which part manages core hardware resources? Answer: kernel. Arrange four layers in order, then reveal the explanation.

THUMBNAIL: KERNEL vs SHELL

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 1: Operating Systems: Structure and Services

Textbook chapter begins at printed page 1 | Selected faceless concepts: 5

CLASS 10 / CH 1

C10-01-03

The process life cycle

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Operating Systems: Structure and Services (p. 1)

HOOK	Why does a running app suddenly wait?
SHORT ANIMATION	Animate tokens moving through new, ready, running, waiting and terminated states.
EXTEND TO 2-5 MIN	Walk through a music-player process and a waiting I/O event.
QUIZ / WEBSITE	Can a waiting process immediately use the CPU? Answer: not until the wait condition clears. Move a process card through states for 5 scenarios.

THUMBNAIL: WHERE DID THE APP GO? A PRIORITY | M EFFORT

CLASS 10 / CH 1

C10-01-04

FCFS CPU scheduling

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Operating Systems: Structure and Services (p. 1)

HOOK	Three apps are waiting. Who goes first?
SHORT ANIMATION	Three arrival cards enter a queue and turn into a simple Gantt strip.
EXTEND TO 2-5 MIN	Solve one original first-come-first-served scheduling example step by step.
QUIZ / WEBSITE	Which of three listed arrival times runs first? Answer: earliest ready arrival. Interactive arrival-order queue and instant Gantt feedback.

THUMBNAIL: WHO GETS THE CPU? B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 1: Operating Systems: Structure and Services

Textbook chapter begins at printed page 1 | Selected faceless concepts: 5

CLASS 10 / CH 1

C10-01-05

RAM and virtual memory

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Operating Systems: Structure and Services (p. 1)

HOOK	Why does a laptop slow down with too many tabs?
SHORT ANIMATION	Show RAM filling with tab cards, then a slower disk-backed overflow illustration.
EXTEND TO 2-5 MIN	Explain memory pressure and virtual memory with a careful simplified analogy.
QUIZ / WEBSITE	Is a disk usually as fast as RAM? Answer: no. Choose tabs to close and watch memory-pressure indicator change.

THUMBNAIL: RAM IS FULL!

A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 2: System Recovery and Advanced Maintenance

Textbook chapter begins at printed page 20 | Selected faceless concepts: 3

CLASS 10 / CH 2

C10-02-01

The PC boot journey

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: System Recovery and Advanced Maintenance (p. 20)

HOOK	What happens before Windows appears?
SHORT ANIMATION	Power button > firmware checks > bootloader > operating system; animated boot timeline.
EXTEND TO 2-5 MIN	Compare the roles of firmware and operating system during startup.
QUIZ / WEBSITE	Does the OS load before the machine begins firmware startup? Answer: no. Put startup cards in chronological order.

THUMBNAIL: BEFORE WINDOWS STARTS

A PRIORITY | M EFFORT

CLASS 10 / CH 2

C10-02-02

Task Manager detective

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: System Recovery and Advanced Maintenance (p. 20)

HOOK	Which app is making your PC slow?
SHORT ANIMATION	Animate CPU, memory and disk bars with one runaway application.
EXTEND TO 2-5 MIN	Explain how to interpret simulated Task Manager / Resource Monitor readings.
QUIZ / WEBSITE	High CPU with low disk: which resource is under most pressure? Answer: CPU. Find the likely culprit in a simulated resource-monitor dashboard.

THUMBNAIL: FIND THE SLOW APP

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 2: System Recovery and Advanced Maintenance

Textbook chapter begins at printed page 20 | Selected faceless concepts: 3

CLASS 10 / CH 2

C10-02-03

Safe Mode versus System Restore

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: System Recovery and Advanced Maintenance (p. 20)

HOOK	Your PC will not start normally. What next?
SHORT ANIMATION	Split screen: troubleshooting route branches into Safe Mode, restore and backup.
EXTEND TO 2-5 MIN	Explain what each recovery approach attempts and its limits; never imply risk-free fixes.
QUIZ / WEBSITE	Which mode starts Windows with minimal drivers? Answer: Safe Mode. Choose recovery options for 3 safe hypothetical scenarios.

THUMBNAIL: PC WON'T START? B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 3: Introduction to Python Programming

Textbook chapter begins at printed page 40 | Selected faceless concepts: 3

CLASS 10 / CH 3

C10-03-01

Variables as labeled boxes

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction to Python Programming (p. 40)

HOOK	What does x = x + 1 really do?
SHORT ANIMATION	A value moves into a labeled box, then changes after an assignment.
EXTEND TO 2-5 MIN	3-minute animated walk-through of assignment and variable updates.
QUIZ / WEBSITE	If x starts at 4 then x = x + 1, what is x? Answer: 5. Predict Python output, run safe sample code and reveal changes.

THUMBNAIL: x = x + 1 ?

A PRIORITY | S EFFORT

CLASS 10 / CH 3

C10-03-02

Python type conversion

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction to Python Programming (p. 40)

HOOK	Why is "5" + 2 an error?
SHORT ANIMATION	Contrast text tile '5' and number tile 2; show conversion before addition.
EXTEND TO 2-5 MIN	Explain strings, integers, basic conversions and common mistakes.
QUIZ / WEBSITE	What does int('5') + 2 produce? Answer: 7. Pick conversions to make 4 broken expressions valid.

THUMBNAIL: TEXT OR NUMBER?

A PRIORITY | S EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 3: Introduction to Python Programming

Textbook chapter begins at printed page 40 | Selected faceless concepts: 3

CLASS 10 / CH 3

C10-03-03

Operator precedence puzzle

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction to Python Programming (p. 40)

HOOK	Will Python say 14 or 20?
SHORT ANIMATION	Animate 2 + 3 * 4 in stages with multiplication highlighted first.
EXTEND TO 2-5 MIN	Explain expression evaluation and parentheses with original examples.
QUIZ / WEBSITE	2 + 3 * 4 equals? Answer: 14. Timed 5-question arithmetic prediction ladder.

THUMBNAIL: 14 OR 20?

A PRIORITY | S EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 4: Control Structures in Python

Textbook chapter begins at printed page 55 | Selected faceless concepts: 4

CLASS 10 / CH 4

C10-04-01

If/else as a decision road

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Control Structures in Python (p. 55)

HOOK	How does code decide who passes?
SHORT ANIMATION	A score enters a decision gate and exits through two labeled paths.
EXTEND TO 2-5 MIN	Build simple if/else and elif branching with grade-like examples.
QUIZ / WEBSITE	If marks = 55 and pass threshold = 50, which branch runs? Answer: pass. Change threshold and watch program path light up.

THUMBNAIL: HOW CODE DECIDES

A PRIORITY | M EFFORT

CLASS 10 / CH 4

C10-04-02

For-loop versus while-loop

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Control Structures in Python (p. 55)

HOOK	Which loop will never stop?
SHORT ANIMATION	Side-by-side counters: fixed-step for loop and condition-driven while loop.
EXTEND TO 2-5 MIN	Show when to choose each loop and a safe termination condition.
QUIZ / WEBSITE	If a condition never becomes false, what may happen? Answer: endless loop. Repair a simulated loop by choosing condition or increment.

THUMBNAIL: STOP THE LOOP!

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 4: Control Structures in Python

Textbook chapter begins at printed page 55 | Selected faceless concepts: 4

CLASS 10 / CH 4

C10-04-03

Break and continue

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Control Structures in Python (p. 55)

HOOK	Can you skip ONE turn without stopping?
SHORT ANIMATION	Moving counter highlights one iteration being skipped versus loop stopping.
EXTEND TO 2-5 MIN	Compare break and continue on a tiny animated sequence.
QUIZ / WEBSITE	Does continue exit the entire loop? Answer: no. Tap where a loop should skip or stop and see execution trace.

THUMBNAIL: BREAK vs CONTINUE

B PRIORITY | S EFFORT

CLASS 10 / CH 4

C10-04-04

Trace a Python output

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Control Structures in Python (p. 55)

HOOK	Can you beat Python in 10 seconds?
SHORT ANIMATION	Reveal three lines of code, trace variable values in a mini table, pause for answer.
EXTEND TO 2-5 MIN	5-minute guided tracing of conditionals and simple loops, without assuming coding experience.
QUIZ / WEBSITE	Use an original loop with two iterations; ask for printed values and reveal. Interactive step-through with variable watch table.

THUMBNAIL: PREDICT THE OUTPUT

A PRIORITY | S EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 5: Data Science

Textbook chapter begins at printed page 69 | Selected faceless concepts: 4

CLASS 10 / CH 5

C10-05-01

Dirty data to clean chart

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Data Science (p. 69)

HOOK	Can a bad spreadsheet fool you?
SHORT ANIMATION	Messy duplicate/missing-value rows become a clean dataset and bar chart.
EXTEND TO 2-5 MIN	Show collect > clean > analyze > visualize on one fictional survey.
QUIZ / WEBSITE	Does a missing value automatically mean zero? Answer: no. Fix 5 broken rows and compare before/after chart.

THUMBNAIL: DIRTY DATA, WRONG ANSWER

A PRIORITY | M EFFORT

CLASS 10 / CH 5

C10-05-02

Structured versus unstructured data

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Data Science (p. 69)

HOOK	Why can a spreadsheet sort names but not photos?
SHORT ANIMATION	Drag rows, images and chat messages into structured and unstructured baskets.
EXTEND TO 2-5 MIN	Explain data types with school attendance versus photos example.
QUIZ / WEBSITE	Is a JPEG image naturally a table of columns? Answer: no. Sort sample items and get contextual feedback.

THUMBNAIL: DATA HAS TWO FACES

A PRIORITY | S EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 5: Data Science

Textbook chapter begins at printed page 69 | Selected faceless concepts: 4

CLASS 10 / CH 5

C10-05-03

How a graph can mislead

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Data Science (p. 69)

HOOK	Same numbers, different story?
SHORT ANIMATION	Show identical data on two bar charts with different y-axis starts.
EXTEND TO 2-5 MIN	Teach graph interpretation and the importance of scales.
QUIZ / WEBSITE	Can truncating a y-axis exaggerate a difference? Answer: yes. Toggle graph scales to judge whether a claim is supported.

THUMBNAIL: THIS GRAPH IS TRICKING YOU A PRIORITY | M EFFORT

CLASS 10 / CH 5

C10-05-04

How SQL finds a result

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Data Science (p. 69)

HOOK	How does a database answer a question?
SHORT ANIMATION	Highlight 20 table rows; WHERE filter keeps matching items.
EXTEND TO 2-5 MIN	Animate basic SELECT / WHERE on a fictional class inventory.
QUIZ / WEBSITE	Which clause filters rows? Answer: WHERE. Build a simple query using dropdowns and see matching records.

THUMBNAIL: FIND IT IN 1 QUERY B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 6: Introduction To Artificial Intelligence (AI), and Machine Learning (ML)

Textbook chapter begins at printed page 88 | Selected faceless concepts: 4

CLASS 10 / CH 6

C10-06-01

AI versus ML

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction To Artificial Intelligence (AI), and Machine Learning (ML) (p. 88)

HOOK	Are AI and machine learning the same thing?
SHORT ANIMATION	Big AI circle and smaller ML circle with familiar examples filling each.
EXTEND TO 2-5 MIN	Differentiate AI and ML without claiming all AI learns from data.
QUIZ / WEBSITE	Is ML a subset of AI? Answer: yes. Sort everyday technologies into AI, ML or neither with explanations.

THUMBNAIL: AI vs ML IN 30 SECONDS A PRIORITY | M EFFORT

CLASS 10 / CH 6

C10-06-02

Teaching a model with examples

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction To Artificial Intelligence (AI), and Machine Learning (ML) (p. 88)

HOOK	Can a computer learn cats versus dogs?
SHORT ANIMATION	Labeled sample photos feed a training stage; new image gets predicted label.
EXTEND TO 2-5 MIN	Introduce supervised learning and why dataset quality matters.
QUIZ / WEBSITE	What helps supervised learning? Answer: labeled examples. Label a tiny fictional dataset and test a pretend model.

THUMBNAIL: TEACH A COMPUTER A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 6: Introduction To Artificial Intelligence (AI), and Machine Learning (ML)

Textbook chapter begins at printed page 88 | Selected faceless concepts: 4

CLASS 10 / CH 6

C10-06-03

Classification versus regression

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction To Artificial Intelligence (AI), and Machine Learning (ML) (p. 88)

HOOK	Will AI guess a CATEGORY or a NUMBER?
SHORT ANIMATION	One model outputs 'spam/not spam'; another predicts a continuous value.
EXTEND TO 2-5 MIN	Side-by-side visual examples and limitations for both tasks.
QUIZ / WEBSITE	Predicting a temperature is classification or regression? Answer: regression. Match 8 real-world prompts to the correct task type.

THUMBNAIL: CATEGORY OR NUMBER? B PRIORITY | M EFFORT

CLASS 10 / CH 6

C10-06-04

Biased training data

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Introduction To Artificial Intelligence (AI), and Machine Learning (ML) (p. 88)

HOOK	Why can a model make unfair mistakes?
SHORT ANIMATION	A lopsided training sample creates uneven performance across groups.
EXTEND TO 2-5 MIN	Explain bias as a data and evaluation problem; emphasize fair, responsible AI.
QUIZ / WEBSITE	Can a nonrepresentative dataset create poor predictions? Answer: yes. Improve a toy dataset balance and see outcomes change.

THUMBNAIL: WHEN AI GETS IT WRONG A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 7: Applications of AI

Textbook chapter begins at printed page 101 | Selected faceless concepts: 3

CLASS 10 / CH 7

C10-07-01

Speech recognition pipeline

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Applications of AI (p. 101)

HOOK	How does a phone hear your words?
SHORT ANIMATION	Audio waveform > features > text prediction > visible captions.
EXTEND TO 2-5 MIN	Show a simplified speech-to-text pipeline and common errors in noisy audio.
QUIZ / WEBSITE	Does speech recognition understand perfectly in noise? Answer: no. Select best transcript from noisy sample clips or text simulations.

THUMBNAIL: HOW PHONES HEAR YOU A PRIORITY | M EFFORT

CLASS 10 / CH 7

C10-07-02

Computer vision

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Applications of AI (p. 101)

HOOK	How does AI 'see' objects?
SHORT ANIMATION	Camera frame splits into regions, then boxes and labels appear.
EXTEND TO 2-5 MIN	Explain visual recognition with a classroom object-identification example.
QUIZ / WEBSITE	Is an AI detection box a guarantee of correctness? Answer: no. Find false positives in a mock detection screen.

THUMBNAIL: HOW AI SEES A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 7: Applications of AI

Textbook chapter begins at printed page 101 | Selected faceless concepts: 3

CLASS 10 / CH 7

C10-07-03

Recommendation system

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Applications of AI (p. 101)

HOOK	Why does your feed know what you watch?
SHORT ANIMATION	Viewing actions turn into preference signals and a ranked video feed.
EXTEND TO 2-5 MIN	Explain a simple recommender at a conceptual level and its tradeoffs.
QUIZ / WEBSITE	Can recent viewing behavior influence suggestions? Answer: yes. Rank recommended learning videos from sample preferences.

THUMBNAIL: WHY THIS VIDEO? B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 10

CHAPTER 8: Digital Entrepreneurship

Textbook chapter begins at printed page 115 | Selected faceless concepts: 2

CLASS 10 / CH 8

C10-08-01

SWOT analysis as four windows

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Digital Entrepreneurship (p. 115)

HOOK	Could your app idea survive a SWOT check?
SHORT ANIMATION	A fictional student app fills strengths, weaknesses, opportunities and threats boxes.
EXTEND TO 2-5 MIN	Explain how to turn SWOT observations into product decisions.
QUIZ / WEBSITE	Is a competitor usually a strength or threat? Answer: threat. Fill an interactive SWOT canvas for a study-planner idea.

THUMBNAIL: 4 BOXES, 1 BUSINESS

B PRIORITY | S EFFORT

CLASS 10 / CH 8

C10-08-02

Customer journey to useful product

BOOK: Computer Science and Entrepreneurship - Class 10 | CHAPTER: Digital Entrepreneurship (p. 115)

HOOK	Build what students NEED, not just what you like?
SHORT ANIMATION	Student problem > survey > prototype > feedback > iteration timeline.
EXTEND TO 2-5 MIN	A mini case: improving a school revision app through user interviews.
QUIZ / WEBSITE	Should you validate a problem before building everything? Answer: usually yes. Choose the next best question in a 3-step customer-discovery path.

THUMBNAIL: YOUR APP IDEA FAILED?

B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 1: Computer Networks

Textbook chapter begins at printed page 1 | Selected faceless concepts: 6

CLASS 12 / CH 1

C12-01-01

How a message travels

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computer Networks (p. 1)

HOOK	What happens when you hit SEND?
SHORT ANIMATION	Phone message fragments into packets; arrows travel through router and destination.
EXTEND TO 2-5 MIN	Show a simplified path through devices, links, addressing and delivery.
QUIZ / WEBSITE	Does a message need some route through a network? Answer: yes. Put packet journey stages in order, then view the route.

THUMBNAIL: YOUR MESSAGE'S JOURNEY A PRIORITY | L EFFORT

CLASS 12 / CH 1

C12-01-02

Network topologies

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computer Networks (p. 1)

HOOK	What if ONE cable breaks?
SHORT ANIMATION	Compare star, bus and ring layouts; visually cut a link to see consequences.
EXTEND TO 2-5 MIN	Explain tradeoffs and failure points for common topologies.
QUIZ / WEBSITE	Which layout usually has a central connecting device? Answer: star. Reconnect draggable devices into requested topology.

THUMBNAIL: ONE CABLE BREAKS... A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 1: Computer Networks

Textbook chapter begins at printed page 1 | Selected faceless concepts: 6

CLASS 12 / CH 1

C12-01-03

OSI layers as a delivery story

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computer Networks (p. 1)

HOOK	Why does networking have 7 layers?
SHORT ANIMATION	An animated parcel passes through labeled conceptual layers and is unpacked.
EXTEND TO 2-5 MIN	Carefully connect familiar web traffic to the seven-layer mental model.
QUIZ / WEBSITE	Which OSI layer handles end-user application services? Answer: application. Drag simplified functions to OSI layer labels.

THUMBNAIL: 7 LAYERS IN 40 SECONDS

A PRIORITY | L EFFORT

CLASS 12 / CH 1

C12-01-04

Switch versus router

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computer Networks (p. 1)

HOOK	These two boxes do DIFFERENT jobs?
SHORT ANIMATION	Draw two home networks; switch handles local traffic, router connects networks.
EXTEND TO 2-5 MIN	Clarify device purposes with accessible examples and diagrams.
QUIZ / WEBSITE	Which device primarily forwards between different IP networks? Answer: router. Choose the correct device for three office-network scenarios.

THUMBNAIL: SWITCH vs ROUTER

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 1: Computer Networks

Textbook chapter begins at printed page 1 | Selected faceless concepts: 6

CLASS 12 / CH 1

C12-01-05

IP addressing visualized

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computer Networks (p. 1)

HOOK	How does data know which computer?
SHORT ANIMATION	Two networks show unique logical addresses and packet destination labels.
EXTEND TO 2-5 MIN	Explain IP addresses and basic network/host ideas without a long subnet lesson.
QUIZ / WEBSITE	Why is a destination address useful? Answer: to route toward the recipient. Match packets to the correct addressed devices.

THUMBNAIL: WHO GETS THE PACKET? B PRIORITY | M EFFORT

CLASS 12 / CH 1

C12-01-06

Firewall as a checkpoint

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computer Networks (p. 1)

HOOK	Who gets through the network gate?
SHORT ANIMATION	Packet cards approach a rule-based barrier and are allowed or blocked.
EXTEND TO 2-5 MIN	Distinguish basic filtering decisions and a firewall's limits.
QUIZ / WEBSITE	Does a firewall guarantee perfect security? Answer: no. Adjust allow/deny rules on a harmless fictional network.

THUMBNAIL: ACCESS DENIED! A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 2: Computational Thinking & Algorithms

Textbook chapter begins at printed page 20 | Selected faceless concepts: 4

CLASS 12 / CH 2

C12-02-01

Decomposition of a big problem

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computational Thinking & Algorithms (p. 20)

HOOK	How do you solve a huge problem?
SHORT ANIMATION	A school timetable problem splits into rooms, teachers, slots and constraints.
EXTEND TO 2-5 MIN	Break one real problem into smaller computational tasks.
QUIZ / WEBSITE	Does decomposition make complex tasks easier to manage? Answer: yes. Drag task cards into sensible subproblems.

THUMBNAIL: TOO BIG? BREAK IT

A PRIORITY | M EFFORT

CLASS 12 / CH 2

C12-02-02

Algorithm to flowchart

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computational Thinking & Algorithms (p. 20)

HOOK	Can you draw a program before coding?
SHORT ANIMATION	A spoken rule turns into start, decision, process and end shapes.
EXTEND TO 2-5 MIN	Show a small algorithm, flowchart and dry run of its result.
QUIZ / WEBSITE	What shape typically indicates a decision? Answer: diamond. Assemble a missing-steps flowchart and test a sample input.

THUMBNAIL: CODE WITHOUT CODING

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 2: Computational Thinking & Algorithms

Textbook chapter begins at printed page 20 | Selected faceless concepts: 4

CLASS 12 / CH 2

C12-02-03

AND, OR and NOT logic

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computational Thinking & Algorithms (p. 20)

HOOK	Why is TRUE + FALSE not ordinary maths?
SHORT ANIMATION	Two light switches animate Boolean AND, OR and NOT outcomes.
EXTEND TO 2-5 MIN	Explain simple propositions and build a truth table visually.
QUIZ / WEBSITE	True AND False equals? Answer: False. Tap switches and complete missing truth-table cells.

THUMBNAIL: TRUE OR FALSE?

A PRIORITY | M EFFORT

CLASS 12 / CH 2

C12-02-04

Logical inference puzzle

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Computational Thinking & Algorithms (p. 20)

HOOK	Can these statements all be true?
SHORT ANIMATION	Premise cards link by arrows to a conclusion; invalid leap flashes red.
EXTEND TO 2-5 MIN	Explain what counts as a justified conclusion in a simple reasoning puzzle.
QUIZ / WEBSITE	If all A are B and X is A, is X B? Answer: yes. Select valid conclusions from 4 mini arguments.

THUMBNAIL: SPOT THE LOGIC TRAP

B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 3: Object Oriented Programming Using Python

Textbook chapter begins at printed page 33 | Selected faceless concepts: 3

CLASS 12 / CH 3

C12-03-01

Class versus object

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Object Oriented Programming Using Python (p. 33)

HOOK	Is a class the same as an object?
SHORT ANIMATION	Blueprint card produces three distinct student objects with attributes.
EXTEND TO 2-5 MIN	Class, instance, methods and attributes demonstrated via Python visuals.
QUIZ / WEBSITE	Is an object an instance of a class? Answer: yes. Edit a class blueprint and instantiate sample objects.

THUMBNAIL: BLUEPRINT vs OBJECT A PRIORITY | M EFFORT

CLASS 12 / CH 3

C12-03-02

Constructor and self

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Object Oriented Programming Using Python (p. 33)

HOOK	When does an object get its values?
SHORT ANIMATION	A new object is created; initial attribute tokens move through __init__.
EXTEND TO 2-5 MIN	Simplify constructor flow and explain what self refers to in examples.
QUIZ / WEBSITE	Which method commonly initializes a new Python object? Answer: __init__. Trace object creation and predict two attributes.

THUMBNAIL: WHAT IS __init__? B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 3: Object Oriented Programming Using Python

Textbook chapter begins at printed page 33 | Selected faceless concepts: 3

CLASS 12 / CH 3

C12-03-03

Inheritance and polymorphism

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Object Oriented Programming Using Python (p. 33)

HOOK	One parent, many behaviors?
SHORT ANIMATION	Parent class branches to child classes; shared and overridden methods highlight.
EXTEND TO 2-5 MIN	Explain inheritance, overriding and polymorphism with a fictional vehicle model.
QUIZ / WEBSITE	Can a child class provide its own implementation of a method? Answer: yes. Choose which method runs for each object type.

THUMBNAIL: ONE CLASS, MANY FORMS

A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 4: Development of Graphical User Interface (GUI)

Textbook chapter begins at printed page 45 | Selected faceless concepts: 3

CLASS 12 / CH 4

C12-04-01

Your first GUI button

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Development of Graphical User Interface (GUI) (p. 45)

HOOK	Can Python build a clickable app?
SHORT ANIMATION	Blank Tkinter window receives label, input and working button with click effect.
EXTEND TO 2-5 MIN	Build a tiny local GUI from components and explain the event loop conceptually.
QUIZ / WEBSITE	What happens only after clicking the button? Answer: its bound callback executes. Use a safe UI builder to place a button and preview its action.

THUMBNAIL: CLICK IT. CODE RUNS. A PRIORITY | M EFFORT

CLASS 12 / CH 4

C12-04-02

Event-driven programming

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Development of Graphical User Interface (GUI) (p. 45)

HOOK	Who tells the app when to react?
SHORT ANIMATION	Mouse click and keypress icons become event messages triggering handlers.
EXTEND TO 2-5 MIN	Connect user events, handlers and visible UI updates using a form.
QUIZ / WEBSITE	Does a button handler need a triggering event? Answer: yes. Connect events to handlers and view response animations.

THUMBNAIL: HOW A BUTTON THINKS A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 4: Development of Graphical User Interface (GUI)

Textbook chapter begins at printed page 45 | Selected faceless concepts: 3

CLASS 12 / CH 4

C12-04-03

Layout manager showdown

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Development of Graphical User Interface (GUI) (p. 45)

HOOK	Why does this button move?
SHORT ANIMATION	Same labels are rearranged with pack, grid and place concept diagrams.
EXTEND TO 2-5 MIN	Show simple visual differences between positioning strategies, carefully scoped.
QUIZ / WEBSITE	Which strategy commonly arranges widgets by rows and columns? Answer: grid. Arrange 6 widgets using a drag-and-drop grid preview.

THUMBNAIL: PACK vs GRID

B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 5: Code Testing and Debugging

Textbook chapter begins at printed page 60 | Selected faceless concepts: 3

CLASS 12 / CH 5

C12-05-01

Syntax vs runtime vs logic errors

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Code Testing and Debugging (p. 60)

HOOK	One bug, THREE different problems?
SHORT ANIMATION	Three code snippets: parser rejects; runtime crashes; output silently wrong.
EXTEND TO 2-5 MIN	Compare the three bug classes with original Python examples.
QUIZ / WEBSITE	A script runs but gives wrong sum: what bug category? Answer: logic. Diagnose error type from symptom cards and receive tips.

THUMBNAIL: 3 TYPES OF BUGS

A PRIORITY | M EFFORT

CLASS 12 / CH 5

C12-05-02

Debugger step-by-step

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Code Testing and Debugging (p. 60)

HOOK	Why is x suddenly 0?
SHORT ANIMATION	Yellow execution line walks over code, with variable-watch panel updating.
EXTEND TO 2-5 MIN	Demonstrate breakpoints, step into/over and variable inspection conceptually.
QUIZ / WEBSITE	What helps pause execution at a chosen line? Answer: breakpoint. Step through a 6-line example and find first wrong value.

THUMBNAIL: CATCH THE BUG LIVE

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 5: Code Testing and Debugging

Textbook chapter begins at printed page 60 | Selected faceless concepts: 3

CLASS 12 / CH 5

C12-05-03

Edge cases and assertions

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Code Testing and Debugging (p. 60)

HOOK	Your code works... until input is empty?
SHORT ANIMATION	A valid test passes, empty-string test exposes missing condition.
EXTEND TO 2-5 MIN	Explain why tests should include boundaries, unusual inputs and expectations.
QUIZ / WEBSITE	Is testing only the usual input enough? Answer: no. Add two edge cases to an existing mini test suite.

THUMBNAIL: THE TEST YOU FORGOT

B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 6: Data and Analysis

Textbook chapter begins at printed page 72 | Selected faceless concepts: 4

CLASS 12 / CH 6

C12-06-01

Raw data to usable dataset

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Data and Analysis (p. 72)

HOOK	Can you train AI on this messy sheet?
SHORT ANIMATION	Missing values and inconsistent labels flash red, then become organized columns.
EXTEND TO 2-5 MIN	Explain preparing structured data for a basic model.
QUIZ / WEBSITE	Can mislabeled rows affect model performance? Answer: yes. Repair a tiny dataset and receive a quality score.

THUMBNAIL: CLEAN DATA FIRST A PRIORITY | M EFFORT

CLASS 12 / CH 6

C12-06-02

Supervised versus unsupervised ML

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Data and Analysis (p. 72)

HOOK	Who gave the model the answers?
SHORT ANIMATION	Labeled apple/orange dataset versus unlabeled dots forming groups.
EXTEND TO 2-5 MIN	Contrast category prediction with pattern discovery using original toy examples.
QUIZ / WEBSITE	Which uses labeled target examples? Answer: supervised learning. Choose appropriate technique for 5 tasks.

THUMBNAIL: ANSWERS OR PATTERNS? B PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 6: Data and Analysis

Textbook chapter begins at printed page 72 | Selected faceless concepts: 4

CLASS 12 / CH 6

C12-06-03

Train versus test split

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Data and Analysis (p. 72)

HOOK	Why shouldn't AI study its exam answers?
SHORT ANIMATION	Dataset cards split into training stack and held-out testing stack.
EXTEND TO 2-5 MIN	Explain generalization and why evaluation on unseen data matters.
QUIZ / WEBSITE	Should test examples be used as ordinary training examples? Answer: no. Pick a data split and compare toy model results.

THUMBNAIL: AI CHEATING ON ITS TEST? A PRIORITY | L EFFORT

CLASS 12 / CH 6

C12-06-04

Precision versus recall

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Data and Analysis (p. 72)

HOOK	Which AI mistake is worse?
SHORT ANIMATION	A detection grid shows false alarms versus misses for two imaginary models.
EXTEND TO 2-5 MIN	Explain precision and recall tradeoffs with clearly labeled confusion outcomes.
QUIZ / WEBSITE	Missing a real positive harms which count? Answer: false negative. Adjust threshold on toy classifier and observe precision/recall shift.

THUMBNAIL: ACCURATE... OR NOT? A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 7: Hypothesis Testing

Textbook chapter begins at printed page 89 | Selected faceless concepts: 3

CLASS 12 / CH 7

C12-07-01

Null versus alternative hypothesis

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Hypothesis Testing (p. 89)

HOOK	Can we prove a claim with one result?
SHORT ANIMATION	Two hypothesis cards meet a small sample of coin flips.
EXTEND TO 2-5 MIN	Introduce H0/H1 as a structured question; avoid proving certainty.
QUIZ / WEBSITE	Does a statistical test typically compare evidence against H0? Answer: yes. Select sensible H0 and H1 for sample school-survey claims.

THUMBNAIL: WHAT IS THE CLAIM? A PRIORITY | M EFFORT

CLASS 12 / CH 7

C12-07-02

Sampling variability

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Hypothesis Testing (p. 89)

HOOK	Ten students said YES. Is that enough?
SHORT ANIMATION	Several random samples animate different bar heights from same population.
EXTEND TO 2-5 MIN	Explain why results vary across samples and why sample choice matters.
QUIZ / WEBSITE	Can two random samples produce different percentages? Answer: yes. Resample a simulated poll and watch estimates change.

THUMBNAIL: 10 PEOPLE = TRUTH? A PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 7: Hypothesis Testing

Textbook chapter begins at printed page 89 | Selected faceless concepts: 3

CLASS 12 / CH 7

C12-07-03

What a p-value does NOT say

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Hypothesis Testing (p. 89)

HOOK	Does p = 0.03 mean a claim is 97% true?
SHORT ANIMATION	A p-value ruler highlights the observed evidence under a null model; myth crossed out.
EXTEND TO 2-5 MIN	Careful conceptual intro to significance and common interpretation errors.
QUIZ / WEBSITE	Is p-value the probability that H0 is true? Answer: no. Choose valid interpretations of short hypothetical reports.

THUMBNAIL: THE p-VALUE MYTH B PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 8: Applications of Computer Science

Textbook chapter begins at printed page 103 | Selected faceless concepts: 3

CLASS 12 / CH 8

C12-08-01

Smart agriculture with IoT

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Applications of Computer Science (p. 103)

HOOK	Can a plant ask for water?
SHORT ANIMATION	Soil sensor reads low moisture and triggers a controller and pump.
EXTEND TO 2-5 MIN	Walk through sensor > network > decision > actuator in a simple farm scenario.
QUIZ / WEBSITE	What measures soil moisture? Answer: a sensor. Tune sensor threshold in a simulated watering system.

THUMBNAIL: PLANTS THAT TEXT YOU? A PRIORITY | L EFFORT

CLASS 12 / CH 8

C12-08-02

How cloud computing feels

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Applications of Computer Science (p. 103)

HOOK	Where is your file when it is 'in the cloud'?
SHORT ANIMATION	Phone uploads to remote service; laptop retrieves same authorized file.
EXTEND TO 2-5 MIN	Explain remote storage/services, availability and access risks.
QUIZ / WEBSITE	Is a cloud-stored file necessarily saved only on your phone? Answer: no. Drag local/cloud resources into correct place in a simple architecture.

THUMBNAIL: WHERE DID MY FILE GO? A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 8: Applications of Computer Science

Textbook chapter begins at printed page 103 | Selected faceless concepts: 3

CLASS 12 / CH 8

C12-08-03

Blockchain as linked records

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Applications of Computer Science (p. 103)

HOOK	What if someone changes an old record?
SHORT ANIMATION	Record blocks show hashes chaining; tampering changes following checks.
EXTEND TO 2-5 MIN	Explain tamper-evidence as a conceptual feature, without promising absolute security.
QUIZ / WEBSITE	Would altering recorded content affect its hash? Answer: generally yes. Edit a toy block's data and watch hashes change.

THUMBNAIL: CHANGE ONE BLOCK...

B PRIORITY | L EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 9: Cybersecurity and Safe Digital Collaboration

Textbook chapter begins at printed page 118 | Selected faceless concepts: 4

CLASS 12 / CH 9

C12-09-01

Spot the phishing message

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Cybersecurity and Safe Digital Collaboration (p. 118)

HOOK	Could you spot this fake school notice?
SHORT ANIMATION	Two near-identical messages appear, with URL, urgency and sender clues zoomed.
EXTEND TO 2-5 MIN	Teach practical phishing red flags; no links to real attacks.
QUIZ / WEBSITE	Is urgent password demand a warning sign? Answer: yes. Spot 4 red flags in an original mock message.

THUMBNAIL: DON'T CLICK THIS!

A PRIORITY | M EFFORT

CLASS 12 / CH 9

C12-09-02

Two-factor authentication

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Cybersecurity and Safe Digital Collaboration (p. 118)

HOOK	Why isn't a password always enough?
SHORT ANIMATION	Password layer opens first lock; second trusted factor opens another.
EXTEND TO 2-5 MIN	Show extra authentication factor and limitations, including phishing-resistance nuance.
QUIZ / WEBSITE	Does 2FA add another verification step? Answer: yes. Choose safer login setups in example scenarios.

THUMBNAIL: PASSWORD IS NOT ENOUGH

A PRIORITY | M EFFORT

BOOK: COMPUTER SCIENCE AND ENTREPRENEURSHIP - CLASS 12

CHAPTER 9: Cybersecurity and Safe Digital Collaboration

Textbook chapter begins at printed page 118 | Selected faceless concepts: 4

CLASS 12 / CH 9

C12-09-03

Ransomware and backups

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Cybersecurity and Safe Digital Collaboration (p. 118)

HOOK	What if your files are suddenly locked?
SHORT ANIMATION	Safe animated attack scenario; local files become inaccessible while offline backup survives.
EXTEND TO 2-5 MIN	Explain prevention and recovery principles; no operational attack details.
QUIZ / WEBSITE	Can isolated, tested backups improve recovery chances? Answer: yes. Plan a simple 3-copy backup strategy in an interactive canvas.

THUMBNAIL: ALL FILES LOCKED?

A PRIORITY | M EFFORT

CLASS 12 / CH 9

C12-09-04

Digital footprint and privacy

BOOK: Computer Science and Entrepreneurship - Class 12 | CHAPTER: Cybersecurity and Safe Digital Collaboration (p. 118)

HOOK	What does the Internet remember about you?
SHORT ANIMATION	Old public posts remain as visible breadcrumbs through later searches.
EXTEND TO 2-5 MIN	Explain what public sharing exposes and basic account privacy decisions.
QUIZ / WEBSITE	Can a public post remain discoverable after being shared by others? Answer: yes. Audit a fictional student profile for privacy risks.

THUMBNAIL: YOUR DIGITAL SHADOW

A PRIORITY | M EFFORT

Students must know the grade and chapter at a glance



Recommended hierarchy

Top-left: CLASS 10 or CLASS 12 + CHAPTER NUMBER. The main 2-5 words: curiosity hook. Smaller metadata in the description: exact full chapter title and textbook. Keep text bold, mobile-sized and consistent.

- Use one source of truth: grade, chapter number and full chapter name from this document.
- Never label a Chapter 1 video only "Chapter 1" without the grade.
- Hook wording can be punchy; do not use fake exam guarantees, fake claims or invented board rankings.
- Use diagrams you created yourself, not screenshots of copyrighted textbook pages.

A realistic solo-creator approach: 2-3 unique topics weekly

Identity + relatable pain

WEEK 1

TOPICS OS controls your apps (10); Spot phishing (12); Variables as boxes (10)

DELIVER 1 face-to-camera introduction; 3 Shorts; 1 optional 2-5m explainer

Animations people want to share

WEEK 2

TOPICS Network topologies (12); AI vs ML (10); Algorithm to flowchart (12)

DELIVER 3 Shorts; 1 2-5m faceless; 3 original website challenges

Interactive prediction

WEEK 3

TOPICS Process life cycle (10); Class vs object (12); For/while loops (10)

DELIVER 3 Shorts; 1 code challenge; 1 micro explainer

Proof that the site helps

WEEK 4

TOPICS Misleading graph (10); First GUI button (12); Speech recognition (10)

DELIVER 3 Shorts; 2 visual explainers; website progress screenshots

Deeper concepts made easy

WEEK 5

TOPICS OSI layers (12); Train/test split (12); Task Manager detective (10)

DELIVER 3 Shorts; 1 long faceless diagram video; quiz landing pages

Bridge to first sequential lecture

WEEK 6

TOPICS IoT smart agriculture (12); Python errors (12); If/else decision road (10)

DELIVER 3 Shorts; announce the first full chapter lecture with a date you can keep

Remotion + Excalidraw + slide animations + website

Name the source correctly

Metadata every time: Class, chapter number, exact chapter name, book title, and textbook chapter opening page. Add an internal source-check flag before publishing.

Make an animation, not a moving bullet list

Prefer tokens moving through a system, before/after comparisons, changed program values, network routing, or an interactive prediction pause. Excalidraw paths and Remotion timelines can make causality visible.

Build one master and two crops

Create original diagrams as vector layers. Keep the important action in a 9:16 safe region. Export a 16:9 master for 2-5 minute YouTube and a deliberate 9:16 Short with large text; do not simply crop the edge of a landscape frame.

Check truth and board relevance

Use the PDFs to verify the topic belongs in that chapter. Independently verify factual examples and original quiz answers. Do not assume every interesting add-on is in the book. Use no official board-weightage claims without current board evidence.

Give every Short a useful ending

Prefer "Try the 3-question check" over an intrusive subscribe plea. Deep link to /class-10/ch-01/your-topic or /class-12/ch-01/your-topic. These URL patterns are suggestions, not existing URLs.

Transition without a dramatic reveal

Use your own voice and brand from day one. Put one simple on-camera introduction between the first 3-6 Shorts. Start full sequential lectures once 1-2 chapters are ready and your output cadence is sustainable.

ORIGINAL / NON-VERBATIM

The titles, scenarios, quiz prompts and thumbnail copy in this guide are new production suggestions. They are not an official exam guide or copied passages from either book.